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SECTION 23 00 00 — HVAC

1. VRF System shall be heat-recovery type with minimum COP of 4.0 at AHRI conditions.
2. All refrigerant piping shall be ACR grade copper, brazed joints, pressure tested at 600 psi.
3. Pre-insulated ductwork (PIR type) for all supply and return ducts.
4. Fresh air handling units with heat recovery wheel, minimum 70% effectiveness.
5. All equipment to comply with ASHRAE 90.1 energy efficiency standards.
6. Commissioning as per ASHRAE Guideline 0-2005.

SECTION 26 00 00 — ELECTRICAL

1. Main LV Switchboard: 4000A, 415V, 3-phase, 4-wire, 50Hz.
2. Sub-distribution boards: MCB/MCCB type, IP54 rated.
3. Cable trays: Hot-dip galvanized, ladder type for power, perforated for data.
4. Lighting: LED throughout, minimum 90 CRI, dimmable in office areas.
5. Emergency lighting: maintained type, 3-hour battery backup.
6. Earthing: 2 × earth pits per riser, interconnected ring.
7. Lightning protection: ESE type, coverage per BS EN 62305.

SECTION 22 00 00 — PLUMBING

1. Water supply: PPR pipes (hot & cold), pressure rated PN20.
2. Drainage: uPVC pipes, BS EN 1401, with anti-siphon traps.
3. Sanitary fixtures: Duravit / Grohe / equivalent, water-saving type.
4. Water tanks: GRP sectional, 2-day storage capacity.
5. Pumps: Grundfos / Wilo, variable speed for boosting.
6. Solar water heating pre-piping for future installation.

SECTION 21 00 00 — FIRE PROTECTION

1. Sprinkler system: wet type, K-factor 5.6, ordinary hazard Group 1.
2. Fire hydrant system: Class II standpipe, 100mm risers.
3. Fire pump: diesel + electric, 100% standby, UL/FM listed.
4. FM200: clean agent for server room (50 sqm), 7% concentration.
5. All components UL/FM listed, installation per NFPA 13, 14, 20.
6. Fire alarm: addressable, Notifier / Honeywell / Edwards.

GENERAL REQUIREMENTS

- All works to comply with Dubai Municipality, DEWA, and Civil Defence regulations.
- Contractor to provide shop drawings, material submittals, and O&M manuals.
- 12-month defects liability period from practical completion.
- Performance bond: 10% of contract value.
- Retention: 10% (5% released at practical completion, 5% at end of DLP).
- Insurance: CAR policy covering full contract value.

